HONG TON

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WORK EXPERIENCE

Department of Computer Science, Troy University **Alabama, US**

Graduate Research Assistant, M.S. in Artificial Intelligence *May 2026 - Present*

- Designed and conducted the research experiments about the **AI-driven network simulation framework** to evaluate the communication performance and smart city resilience under the disaster and infrastructure failure scenarios
- Developed and integrated **the OMNeT++/INET network module**, leveraging SDN and URLLC for low-latency communication; applied **ObsPy** for seismic data analytics to analyze the network failures, packet loss, throughput, delay, and cascading failure
- Contributed to and published research on **the AI, disaster-resilient communications, and propagation failures**

Computer Science Tutor and Lab Assistant *Jan 2026 - May 2026*

- Tutored students in DSA, programming languages **(C++/Python)**, and the foundation of mathematics, such as **Calculus, Algebra, Statistics, and Probability**
- Collaborated with instructors to maintain an efficient computer, such as I/O devices and operating systems in the computer lab environment

Youth Business Association Danang **Da Nang City, Vietnam**

Software Intern & IT Specialist *Jun 2023 – Mar 2024*

- Developed a web/mobile application with **Express.js** for managing participant, donation, and financial data for **600+ enterprise members** in Da Nang city, Vietnam; implemented data management workflows and an admin dashboard

EDUCATION | OPT Work Authorization

Troy University | Transfer Student

- M.S. in Artificial Intelligence - Computer Science | GPA: **4.00/4.00** *Jan 2026 - Expected May 2027*
- B.S. in Computer Science | Magna Cum Laude | GPA: **3.61/4.00** *Graduated Fall 2025*
- $8,500 - $9,000 **Merit Scholarship**; Academic Honors: **Chancellor's List | Provost's List**

Vice President of the Vietnamese Student Association

President of the 5 Good Merits Student Club

Member of the Executive Committee of the Vietnamese Students' Association of DTU University

Certifications: Google AI Essentials | AWS Generative AI | AWS Prompt Engineering

PROJECTS

Department of Computer Science, Troy University **Alabama, US**

D2PSim: A Simulator-Agnostic Disaster-to-Packet Simulation Framework

- Designed and implemented a **multi-layer OMNet++ simulation framework** about seismic hazard propagation, probabilistic infrastructure fragility, and cascading failure with **6,000+ nodes IoT network topology**
- Built a **HAZUS-based probabilistic fragility framework** to assess the structural and lifeline failure risk across **25,000+ building nodes** using OpenStreetMap and USGS ShakeMap data sources
- Evaluated network performance degradation such as **packet delivery ratio, throughput, and end-to-end delay** across **690 earthquake scenarios**

SKILLS AND HOBBIES

- Programming Languages: C/C++ | Python | Java | SQL | JavaScript
- AI/ML: PyTorch | TensorFlow | Keras | Scikit-learn | LLMs (Transformers) | Hugging Face | LangChain |
- Developer & AI Tools: Google Colab | Claude Code | Gemini API | VS Code SSH | Figma
- Web Development: Spring Boot | FastAPI | Flask | React | Next.js | Docker | REST APIs
- Interests: Piano (5+ years) | Guitar (8+ years) | Chess (Elo 800 - top 25%)